

# Khadeeja

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## SUMMARY

Data Science postgraduate with experience building and evaluating LLM and multi-agent AI systems using LangChain, LangGraph, and RAG pipelines. Research experience at University of Galway and industry experience validating production document extraction pipelines.

## EDUCATION

### Aligarh Muslim University

Master of Science (Data Science), CGPA: 7.8

Aligarh, India

Aug 2024 – May 2026

## EXPERIENCE

### AI Intern

Jul 2026 – Present

Star Software

Bhopal, India

- Annotated multi-class datasets (tables, key-value pairs, entity regions) across 500+ industrial documents to support OCR and NER model training.
- Conducted evaluation testing of a multimodal LLM-based document extraction pipeline across extraction and validation stages, including comparative benchmarking between LLM versions (GPT-4.1-mini vs. GPT-5.6 Luna).
- Tested a separate LlamaParse-based extraction pipeline for MTR certificates and a downstream validation layer, documenting field-level failure patterns and flag/fix accuracy.

### Research Intern

May 2026 – Jul 2026

Insight Centre for Data Analytics, University of Galway, Ireland

Supervisor: Prof. Saeed H. Alsamhi

- Architected MAAI-SkinDx, a 7-agent LangGraph pipeline for automated dermatological diagnosis using a ReAct protocol with shared episodic memory.
- Benchmarked 5 foundation models and selected ResNet-50 on precision-recall tradeoff, reducing inference latency by 18%.

## SKILLS

**GenAI & Agentic AI:** LangChain, LangGraph, RAG Pipelines, Multi-Agent Systems, Prompt Engineering, Sentence Transformers, Vector Databases, Vision-Language Models (VLMs), BM25 Retrieval, FAISS, LlamaParse

**Machine Learning & Deep Learning:** Scikit-Learn, TensorFlow, Keras, PyTorch, XGBoost, BERT, Hugging Face Transformers, CNN, LSTM, GRU, ResNet-50, ViT, GNN, Multimodal Deep Learning, PCA, SMOTE, Cross-Validation

**Data Analysis & Visualisation:** Pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI, Feature Engineering, ETL Pipelines, SHAP

**Deployment & Interfaces:** FastAPI, Flask, Docker, Streamlit, Gradio, Hugging Face Spaces

**Languages & Tools:** Python, SQL, PostgreSQL, MySQL, Git, GitHub

## PROJECTS

### Voice to SQL Assistant 🔗

Present

Python | OpenAI Whisper | LangChain | Groq LLaMA | PostgreSQL | Streamlit | FastAPI

- End-to-end voice-to-SQL system enabling non-technical users to query enterprise databases without writing SQL; transcribes voice via Whisper, converts to SQL using LangChain + Groq LLaMA, executes on PostgreSQL, returns plain English answers.
- Implemented SQL guardrails preventing injection attacks; deployed on Hugging Face Spaces with Streamlit frontend and FastAPI backend for real-time inference. (*In Progress*)

### SHL Assessment Recommender – Conversational AI Agent 🔗

2026

FastAPI | LangChain | Groq LLaMA 3.3 70B | BM25 | TF-IDF | Docker | Hugging Face Spaces

- Built a conversational AI agent recommending SHL assessments from a catalog of 377 items using hybrid BM25 + TF-IDF retrieval and Groq LLaMA 3.3 70B with enforced JSON output mode – achieving 100% schema compliance across all evaluation traces.
- Implemented 4 conversational behaviors (clarify, recommend, refine, compare) with few-shot prompting, URL validation guardrails, and off-topic refusal; deployed via Docker on Hugging Face Spaces with a FastAPI backend and interactive chat UI.

### Multimodal Fake Review Detection System 🔗

2026

BERT | ViT | TensorFlow | Cross-Modal Attention | GNN | Hugging Face Spaces

- Developed end-to-end multimodal deep learning system using BERT (text) + Vision Transformer (image) achieving 99.72% F1-score on 12,254 samples; implemented novel Cross-Modal Attention architecture for detecting text-image inconsistencies in fraudulent reviews.
- Engineered feature extraction pipeline fusing 768-dim BERT + 768-dim ViT embeddings; SHAP analysis showed near-equal modality importance (48.7% text, 51.3% image) and deployed live on Hugging Face Spaces for real-time inference.